Roll No.: DL.AI.U4AID24140

Amrita Vishwa Vidyapeetham

School of AI, Faridabad

Mid-Term Examinations, Oct 2024

First Semester

B.Tech AI & Data Science

23MAT106 MATHEMATICS FOR INTELLIGENT SYSTEMS - I

Duration: Two Hours | Maximum: 50 Marks

CO	Course Outcomes
CO01	Apply the fundamental concepts of linear algebra and calculus to solve canonical problems analytically and computationally
CO02	Model and simulate simple physical systems using ordinary differential equations
CO03	Apply the concept of probability and random variables to solve elementary problem

Answer all questions

1. Formulate the given set of linear equations consisting of four unknowns (x, y, z, u) as a single equation in Matrix algebra. (Just Formulate, no need to solve the problem). **[3]**[CO01][BTL3]

$$x + y - 5 = 0$$
$$y + 3x = z + 25$$
$$z + 4y = 15$$
$$u + y = 21 + x$$

2. Can the following matrix equation provide a unique solution for the scalar unknowns a *and* b? Justify your answer based on fundamental principle.

[3][CO01][BTL2]

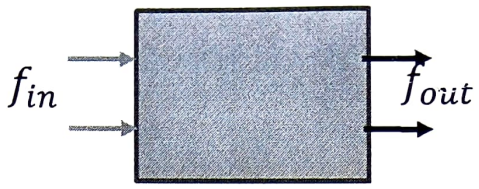
$$(a + b) \times \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} = \begin{bmatrix} 15 & 30 \\ 75 & 60 \end{bmatrix}$$

3. Can the following equation provide a unique solution for the scalar unknowns a, b, c *and* d? Justify your answer based on fundamental principle.

[4][CO01][BTL3]

$$\begin{bmatrix} a & c \\ 2b & 3d \end{bmatrix} + \begin{bmatrix} b & d \\ 2a & 3c \end{bmatrix} = \begin{bmatrix} 15 & 30 \\ 30 & 90 \end{bmatrix}$$

4. An engineering system, shown below, has two input pins and two output pins.
[10][CO01][BTL4]



(a) Formulate a general relationship between the outputs and inputs in Matrix algebra using the concept of System Matrix, clearly mentioning the size of each matrix and vector involved.

(b) If the measured set of output and input vectors are:

input of $\begin{bmatrix}\mathbf{1}\\\mathbf{0}\end{bmatrix}$ gives $\begin{bmatrix}\mathbf{10}\\\mathbf{0}\end{bmatrix}$ as output;

input of $\begin{bmatrix}\mathbf{0}\\\mathbf{1}\end{bmatrix}$ gives $\begin{bmatrix}\mathbf{0}\\\mathbf{20}\end{bmatrix}$ as output;

input of $\begin{bmatrix}\mathbf{1}\\\mathbf{1}\end{bmatrix}$ gives $\begin{bmatrix}\mathbf{10}\\\mathbf{20}\end{bmatrix}$ as output,

determine the system matrix of the system.

(c) Determine the output vector for the input vector $\begin{bmatrix}\mathbf{1}\\\mathbf{2}\end{bmatrix}$

5. Inverse problems involve estimating true object parameters from the system matrix and the measured data.
[10][CO01][BTL4]

(a) Determine the Determinant of the following matrix.

$$\begin{bmatrix}1 & 4 & 6\\2 & 3 & 4\\-1.5 & -6 & -9\end{bmatrix}$$

(b) Solve for the scalar unknowns $\boldsymbol{a, b, c}$ in the following equation using the tools of Matrix algebra. Justify each step.

$$\begin{bmatrix}1 & 4 & 6\\2 & 3 & 4\\-1.5 & -6 & -9\end{bmatrix} \times \begin{bmatrix}a\\b\\c\end{bmatrix} = \begin{bmatrix}5\\4\\-7.5\end{bmatrix}$$

3×3 3×1 3×1

6. Consider the experiment of throwing a dice, and the variable $\boldsymbol{X}$ denotes the outcome of the experiment.
[10][CO03][BTL5]

(a) Determine the complete set of $\boldsymbol{X}$.

(b) Is $\boldsymbol{X}$ a random variable? Justify your answer.

(c) Determine the probability of $\boldsymbol{X \neq 0}$.

(d) Determine $\boldsymbol{E(X^2)}$, which is the expectation value of $\boldsymbol{X^2}$.

7. A class of 1000 students take a MCQ (multiple choice question) based Quiz consisting of one question. There are 5 options, of which only one is correct. Correct answer gets you 10 marks, and wrong answer gets you 0 marks. If all the students **randomly** answer the question, determine the following measures of the marks in the class? **[10]**[CO03][BTL5]
   (a) Average
   (b) Standard Deviation

***

Useful Information

$$1^2 + 2^2 + \cdots + N^2 = \frac{N(N+1)(2N+1)}{6}$$

Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**30**	BTL 1	**0**
CO02	**0**	BTL 2	**3**
CO03	**20**	BTL 3	**7**
		BTL 4	**20**
		BTL 5	**20**
		BTL 6	**0**